

From Model to Prototype

Lecture 04
ITP4256 AI Application Studies

Today's focus & Takeaways

By the end of this lecture, you should be able to:

1. Explain why **Model + Interface + Evaluation** = a real prototype (and what's missing if you only have one or two).
2. Describe the **AI-assistant loop**: Prompt → Code → Test → Fix.
3. **Guardrails** for using AI tools honestly, and why each one exists.
4. Explain why a prototype needs **other platforms** — and that more than one tool can do each job.

Driving question



CA1 model produced an Accuracy and a Recall number, printed inside a Colab notebook.



If a hospital doctor — who has never opened Colab — asked to "try it", what would you show them?

From Prediction to Product

The Decision Loop



Today's question: how does a human actually REACH that last step?

- A print() statement in your notebook is not something a manager can click. Somewhere between "Prediction" and "Judgment", **someone has to build an interface.**

What is a Prototype?

A prototype that people can actually try = **3 things**, all present:



Part	Metaphor	What it means
Model	The brain	A real model, trained on real data
Interface	The face	Something a non-coder can click / type into (e.g. Streamlit, Gradio)
Evaluation	The trust	You can say <i>how</i> good it is and <i>where</i> it fails

- **Missing the Model** (UI with fake if/else guesses) = a lie dressed up as a product.
- **Missing the Interface** (a notebook only you can run) = only useful to you.
- **Missing Evaluation** = you cannot tell anyone if it's safe to trust.

You already use AI. Code is a different game

An essay can sound right. Code has a referee: Run.

If it crashes, the answer failed — no matter how confident the AI sounded.

Three fails you have probably already seen:

1. Invents a function / library that **does not exist**
2. Uses column names that are **not in your table**
3. Forgets **empty cells** → `fit()` / `predict()` crashes

So you are the **supervisor**, not the photocopier:

AI for text	AI for code
“Does this read OK?”	“Does this run on my data ?”
You are often the only checker	The error message is a checker
Easy to paste and hope	A red traceback is public



Rules for using AI tools honestly

4 Rules — same for ChatGPT, Gemini, Claude, Copilot, ...

#	Rule	Why
1	Real data only	No inventing numbers to make the demo look good
2	Log it	Every prompt + bug + fix is part of the work — not extra paperwork
3	Real pipeline underneath	Your UI must call a real trained model — never a hardcoded guess dressed up as AI
4	Explain it	Stakeholders may point to any line of YOUR code and ask what it does.



Deploying: three jobs, several choices

A prototype needs an address — Colab is not one

Closing the notebook tab kills the session. To let someone else click, you usually need **three jobs**:

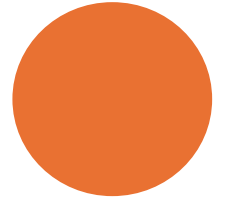
[1. Interface]

+

[2. Store the files]

+

[3. Host a public link]



Deploying: three jobs, several choices (Cont.)

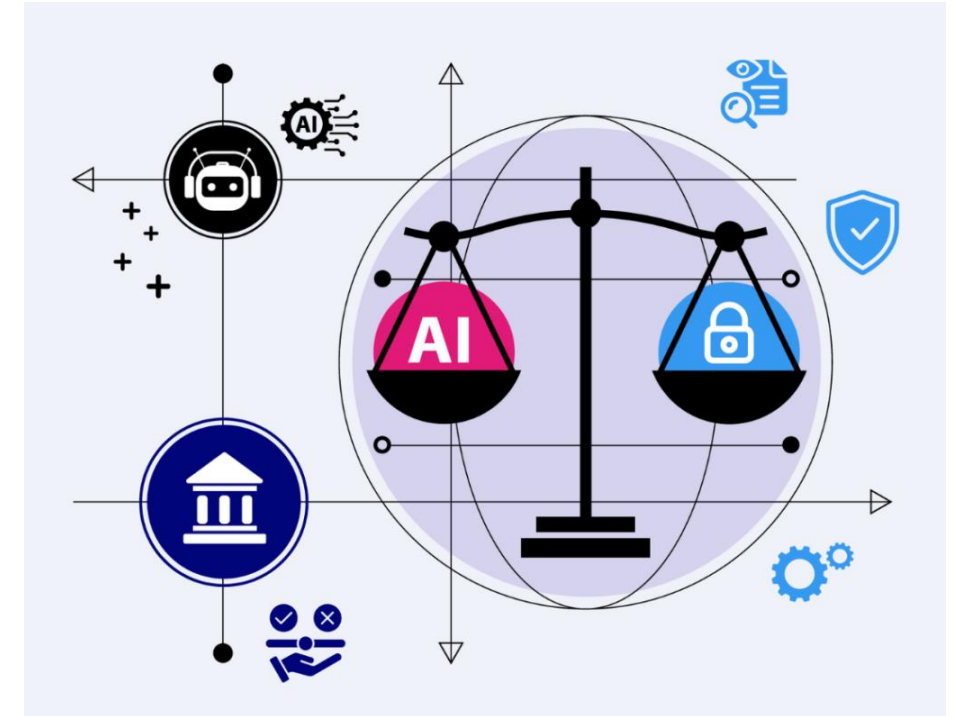
Job	What it does	Common choices (examples)
1. Interface	Turns your Python into a page with sliders / boxes	Streamlit, Gradio
2. Store files	Keeps app + data where a host can read them	GitHub, GitLab
3. Host / share	Gives a URL that still works tomorrow	Streamlit Community Cloud, Hugging Face Spaces, Render, Railway

- You do **not** need every logo on this slide. One working combination is enough — but **GitHub + Streamlit Cloud is not the only path.**
- Free plans and rules **change**. Check the site before you pick.

AI Ethics: Bias, Black Box & Accountability

3 Questions Before You Ship Any Prototype

1. **Bias:** Could your training data under-represent some group (age, gender, region, cheaper cases...)? Would your prototype be less fair to them?
2. **Black Box:** A predicted number alone is not an explanation — can you say *why* the model output this?
3. **Accountability:** If your prototype gives bad advice, who is responsible — the AI, you, or the organisation that deployed it?



Think: if the training records all came from **one source**, would the prototype be fair for everyone you claim to serve?

Putting it together

Same hospital doctor, now with the 4 parts

Go back to Slide 3. A notebook print() is not a prototype. A prototype would need:



Part	Notebook only	Prototype
Model	Trained, sits in Colab	Still a real trained model — not fake rules
Interface	You run cells	The manager clicks / types, no Colab
Evaluation	A score in the output	You can say how good it is and where it fails
Address	Gone when the tab closes	A public link (any host that still works later)

If any row is missing, it is not ready for that manager.